

Ceramics Processing

Presented by Dr Adijat 'Wumi Inyang

Aims of session

&

Learning Outcome



Briefly explain common methods used in the fabrication of ceramics



Explain how the thermal properties of ceramic influence glass-forming techniques



Describe the process that occurs during drying and firing of clay-based ceramics

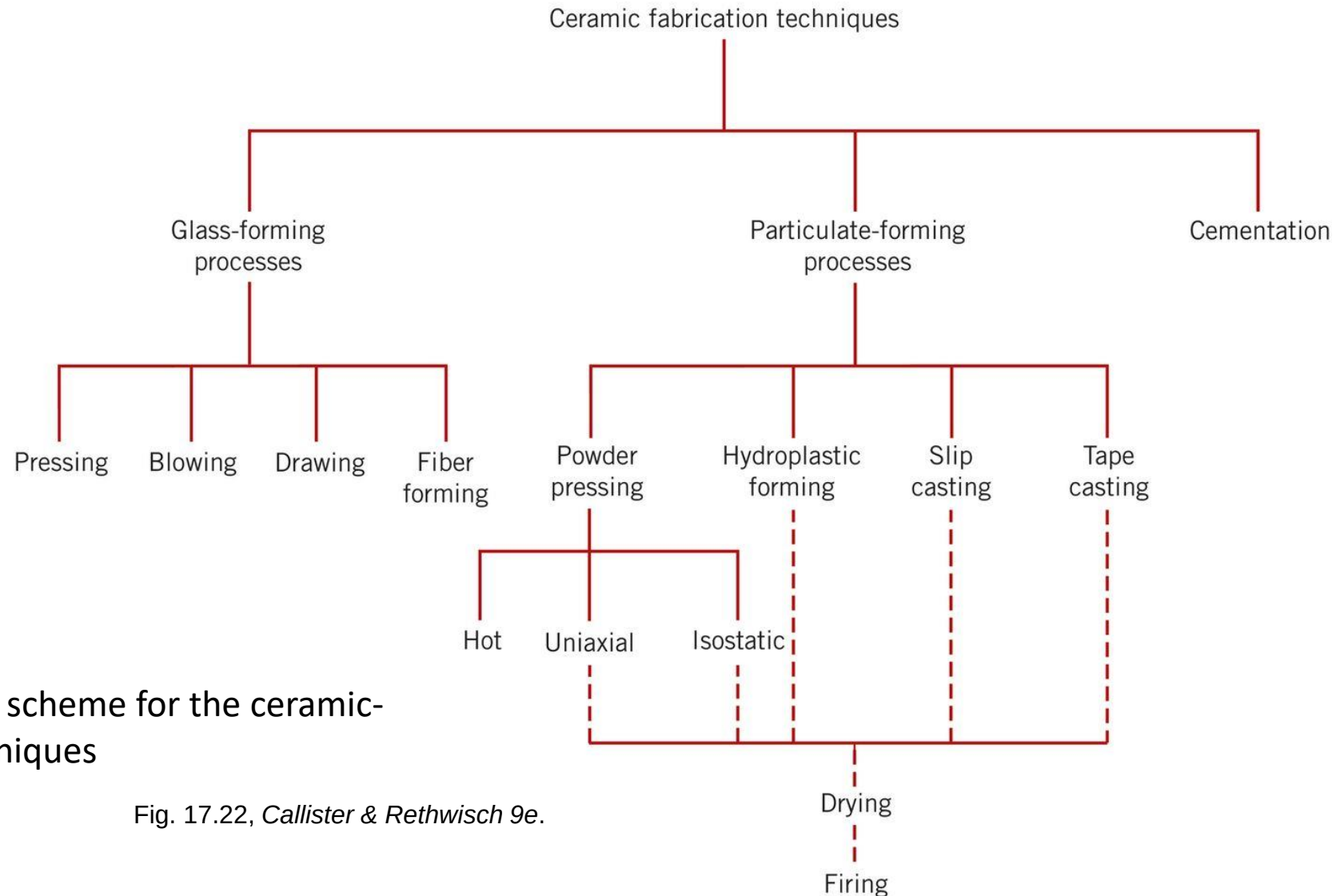


Name and describe forming methods used to fabricate glass pieces



Make diagrams representing the process of clay fabrication

Ceramic processing



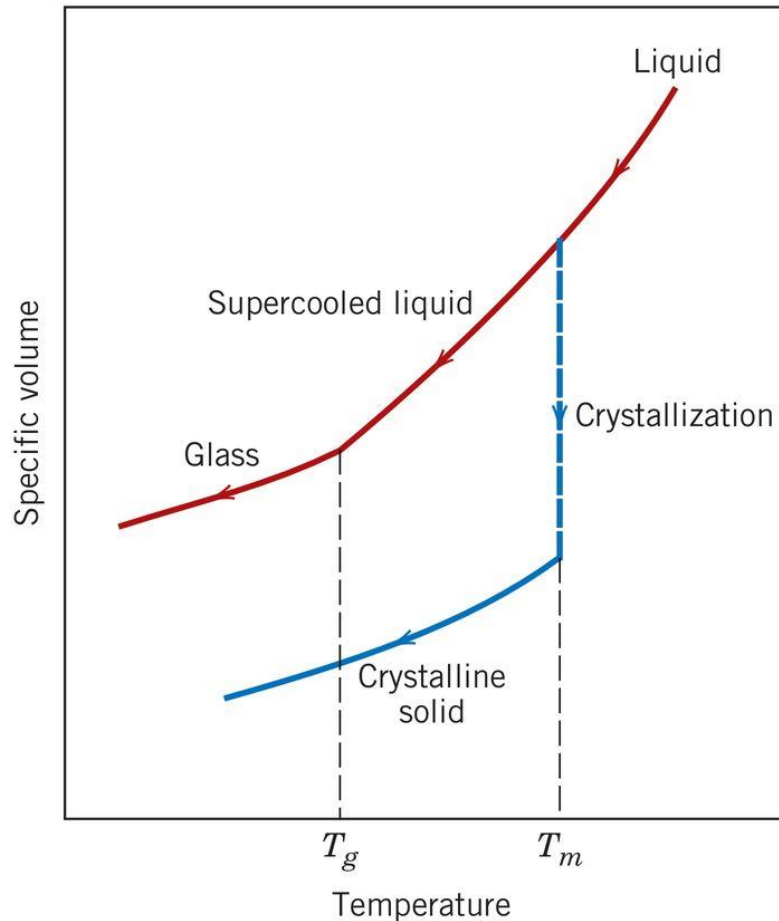
Classification scheme for the ceramic-forming techniques

Fig. 17.22, Callister & Rethwisch 9e.

Glass forming processes



Glass Properties that influence forming methods – Specific volume



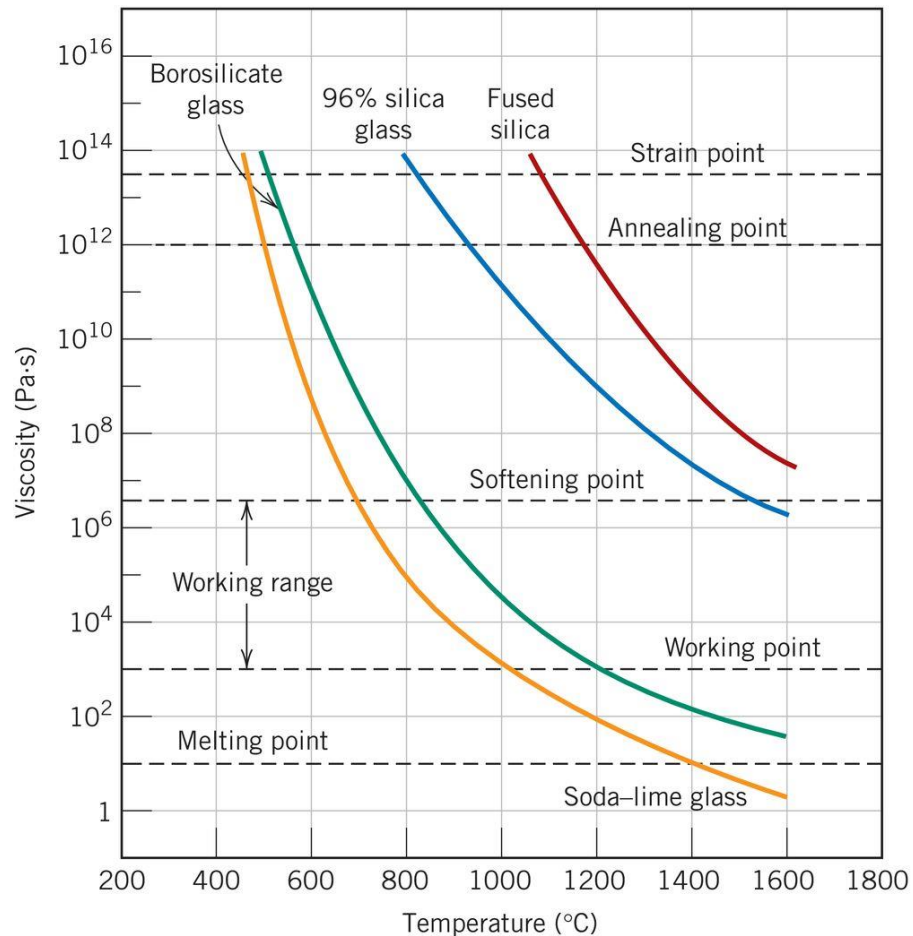
Contrast of specific volume–versus–temperature behavior of crystalline and noncrystalline materials.

- Specific volume ($1/\rho$)
- Crystalline materials:
 - crystallize at melting temp, T_m
 - have abrupt change in spec. vol. at T_m
- Glasses:
 - do not crystallize
 - change in slope in spec. vol. curve at glass transition temperature, T_g
 - transparent - no grain boundaries to scatter light

Glass forming processes



Glass Properties that influence forming methods– Viscosity



From E. B. Shand, Engineering Glass, Modern Materials, Vol. 6, Academic Press, New York, 1968, p. 262.

Fig. 17.24, Callister & Rethwisch 9e.

- Viscosity decreases with T

- soda-lime glass: 70% SiO₂
balance Na₂O (soda) & CaO (lime)
- borosilicate (Pyrex):
13% B₂O₃, 3.5% Na₂O, 2.5% Al₂O₃
- Vycor: 96% SiO₂, 4% B₂O₃
- fused silica: > 99.5 wt% SiO₂

➡ Forming temperature depends on viscosity

Logarithm of viscosity versus temperature for fused silica and three silica glasses

Ceramics Fabrication Methods

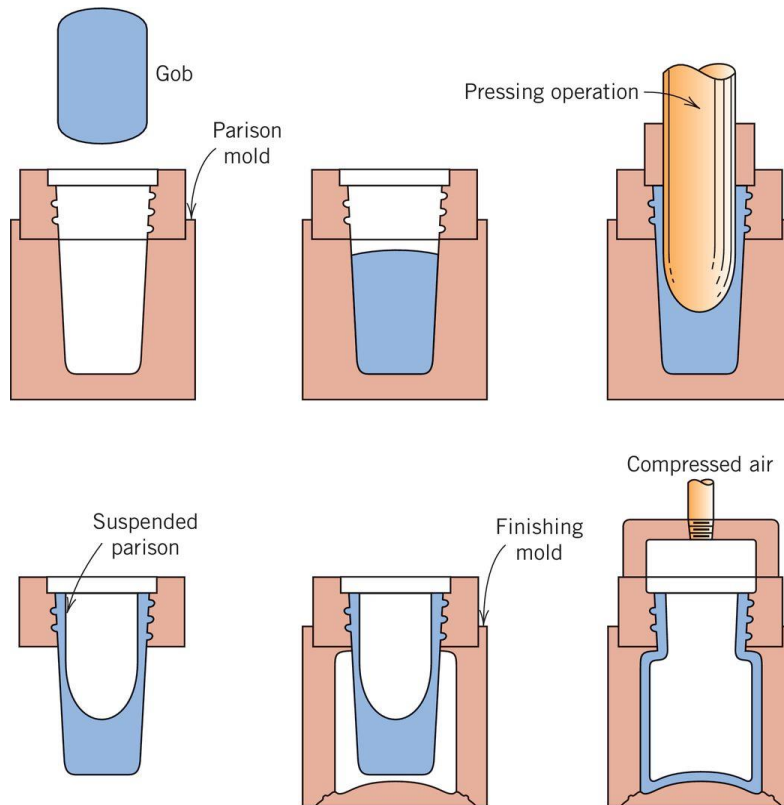


GLASS FORMING

PARTICULATE FORMING

CEMENTATION

- **Blowing of Glass Bottles:**



Adapted from C. J. Phillips, Glass: The Miracle Maker. Reproduced by permission of Pitman Publishing Ltd., London.

- **Pressing:** plates, cheap glasses

- glass formed by application of pressure
- mold is steel with graphite lining

- **Fiber drawing:**

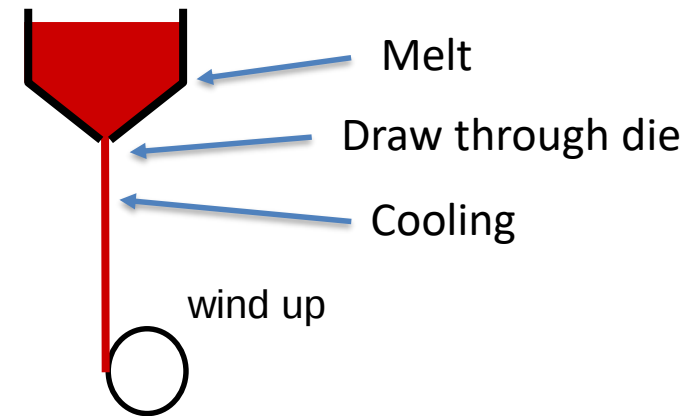
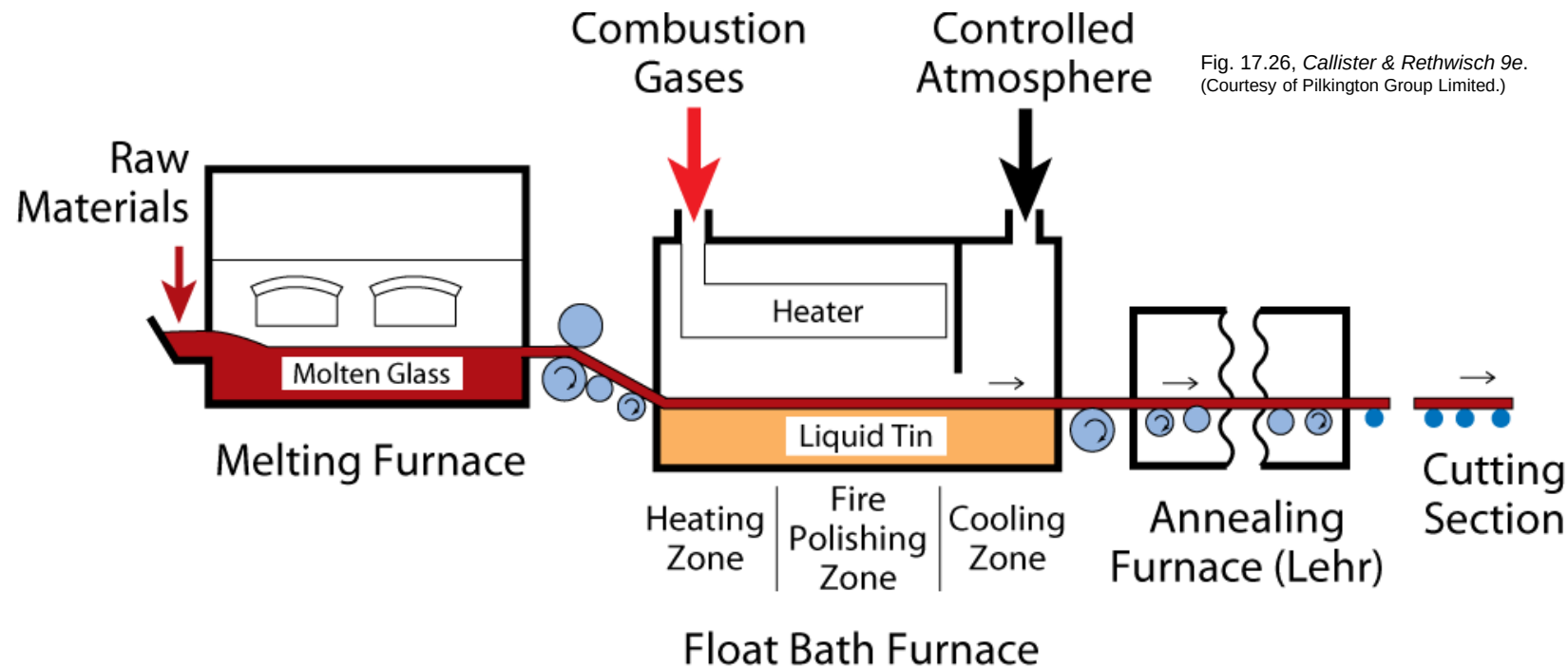


Fig. 17.25, Callister & Rethwisch 9e.

Glass forming processes

Sheet Glass Forming



Float glass process:

- Sheet forming – continuous casting
- Sheets are formed by floating the molten glass on a pool of molten tin
- Perfectly flat glass formed

Glass forming processes

Heat Treating Glass

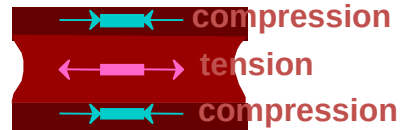
before cooling



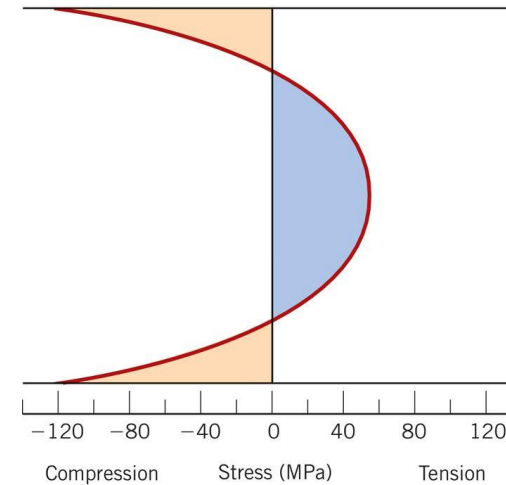
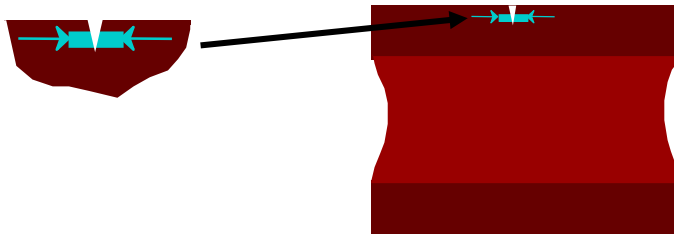
initial cooling



at room temp.



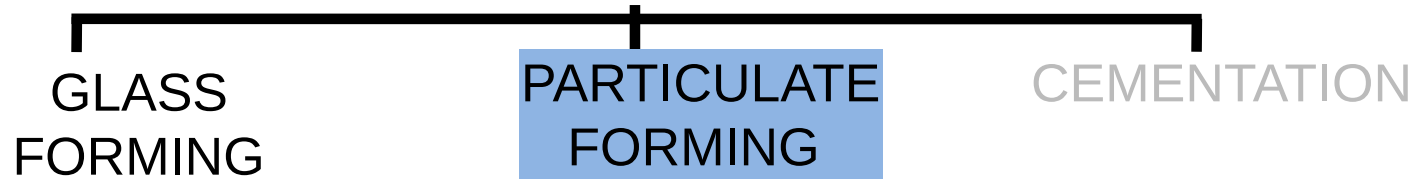
-- Result: surface crack growth is suppressed.



Residual stress profile

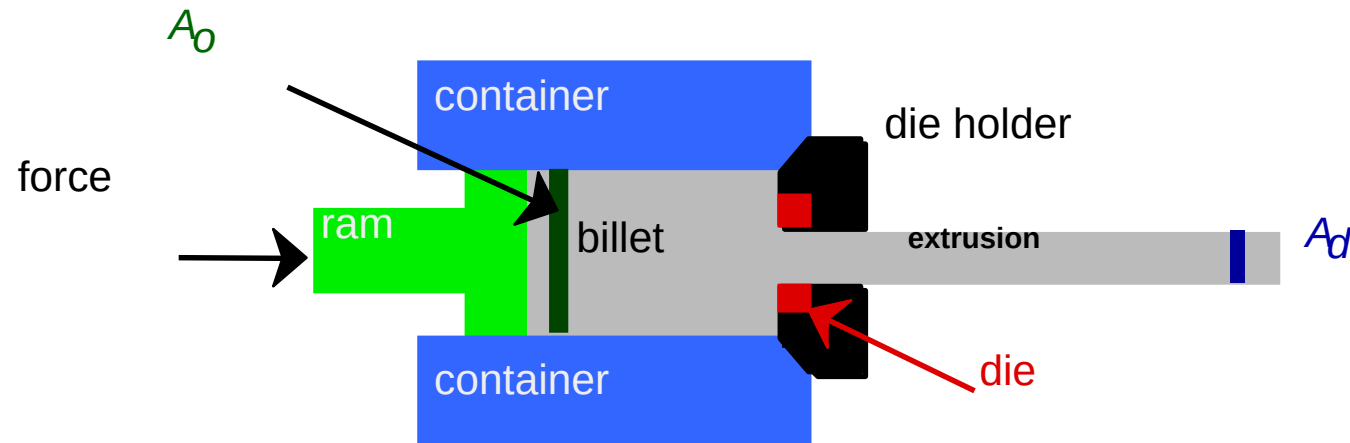
- **Annealing:** removes internal stresses caused by uneven cooling
- **Tempering:**
 - puts surface of glass part into compression
 - suppresses growth of cracks from surface scratches.

Ceramics Fabrication Methods



Hydroplastic forming – clay materials

- Mill (grind) and screen constituents: desired particle size
- Extrude this mass (e.g., into a brick)

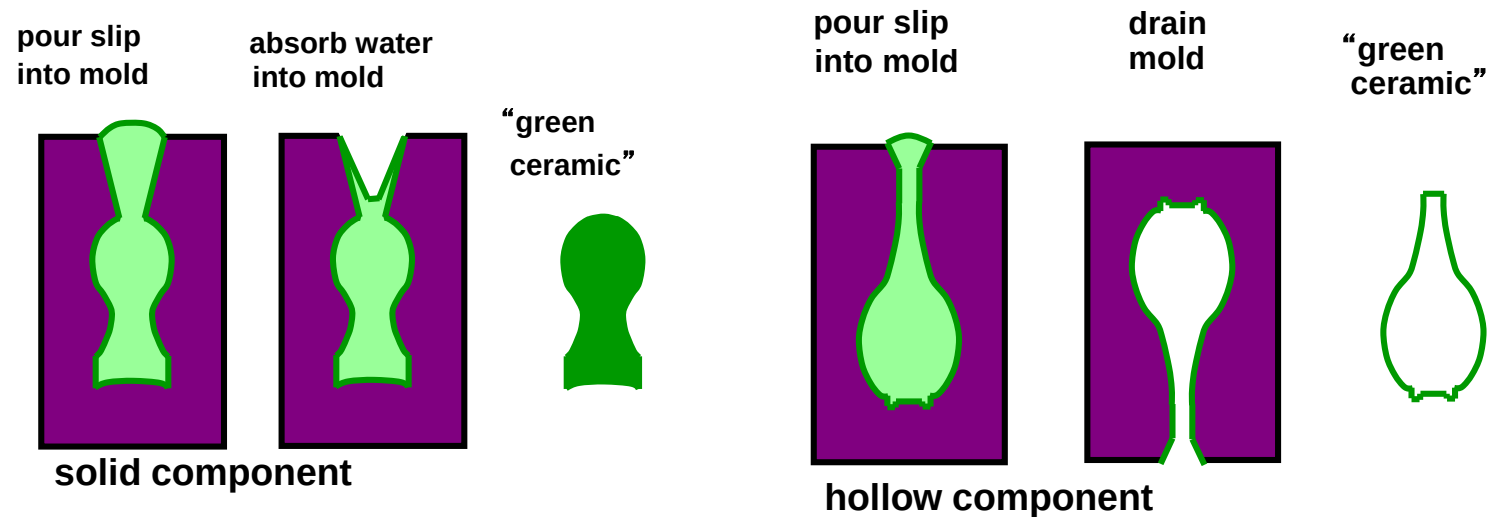


- **Dry** and **fire** the formed piece
- Fire means to dry out and then sinter the particles together

Particulate forming processes

Slip casting

- Mill (grind) and screen constituents: desired particle size
- Mix with water and other constituents to form **slip**
- Slip casting operation



- **Dry** and **fire** the cast piece
- **Green ceramic** is one that is not yet sintered to a solid

Particulate forming processes

Drying and Firing

- **Drying:** as water is removed - interparticle spacing decreases – shrinkage .

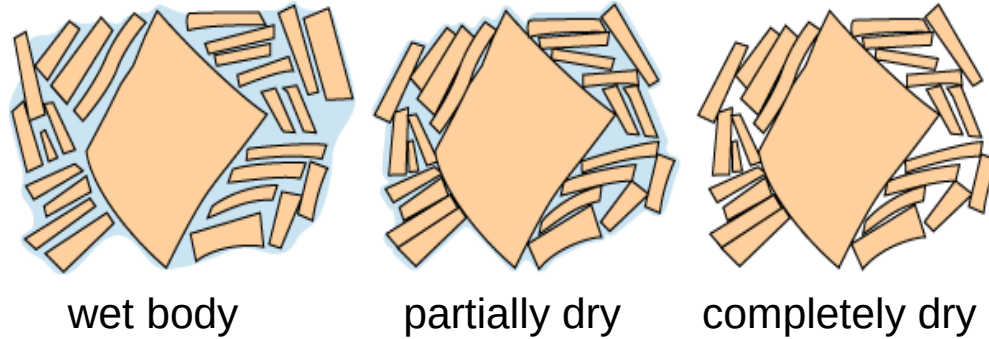


Fig. 17.30, Callister & Rethwisch 9e.

Drying too fast causes sample to warp or crack due to non-uniform shrinkage

- **Firing:**
heat treatment between
900-1400° C
vitrification: liquid glass forms
from clay and flux – flows
between SiO₂ particles.
(Flux lowers melting temperature).

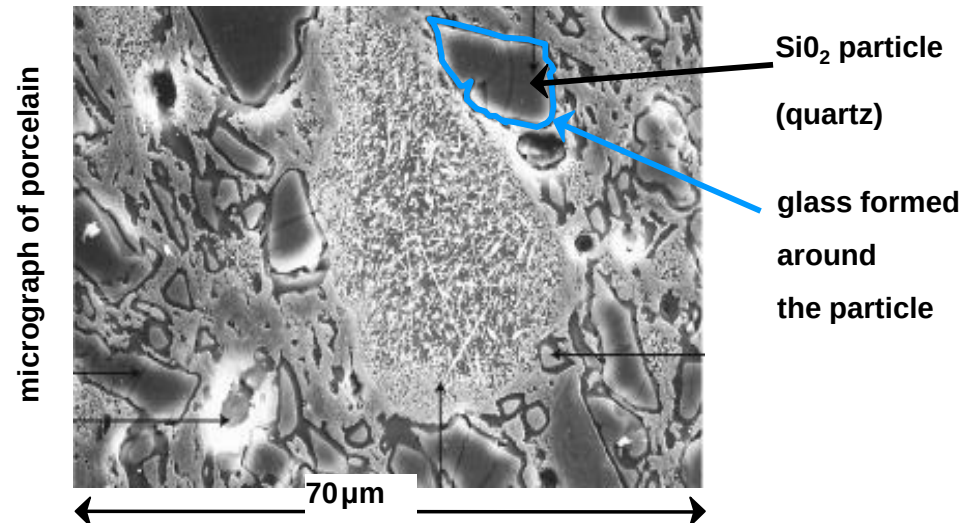


Fig. 17.31, Callister & Rethwisch 9e.

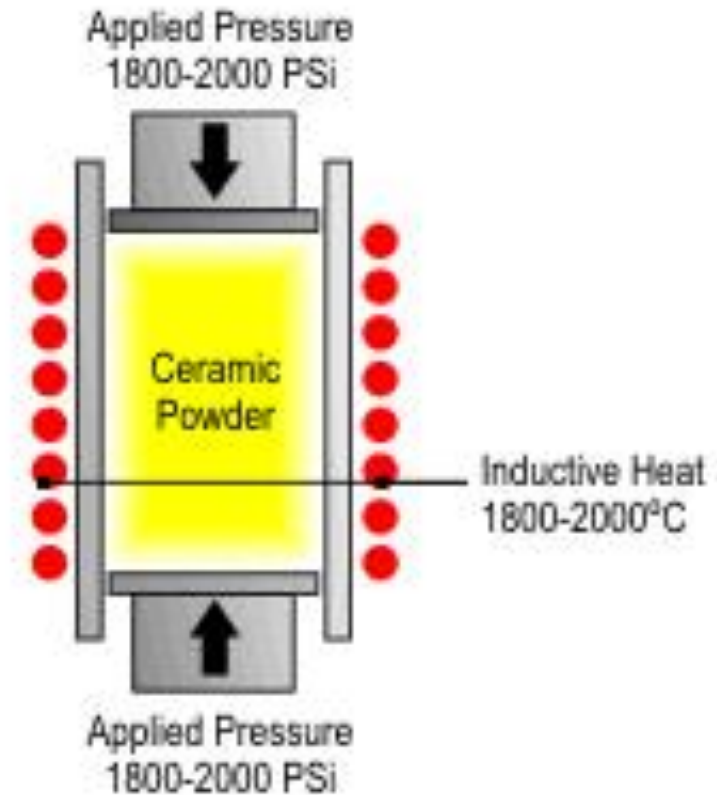
Particulate forming processes



Engineering ceramics

Powder Pressing: used for both clay and non-clay compositions.

- Powder (plus binder) compacted by pressure in a mold
- Uniaxial compression - compacted in single direction
- Isostatic (hydrostatic) compression - pressure applied by fluid - powder in rubber envelope
- Hot pressing - pressure + heat



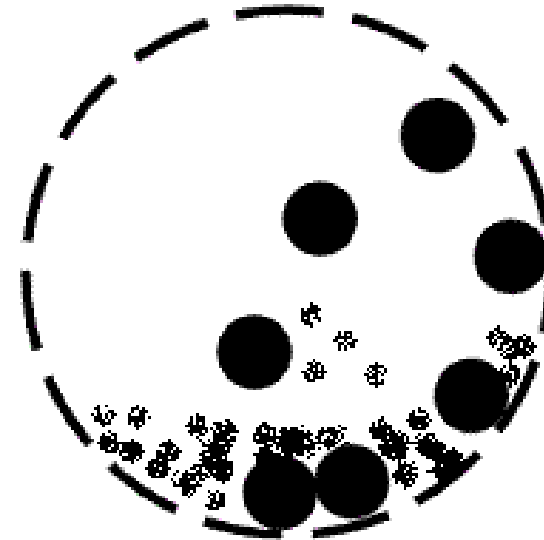
Hot isostatic pressing (HIP)

Particulate forming processes

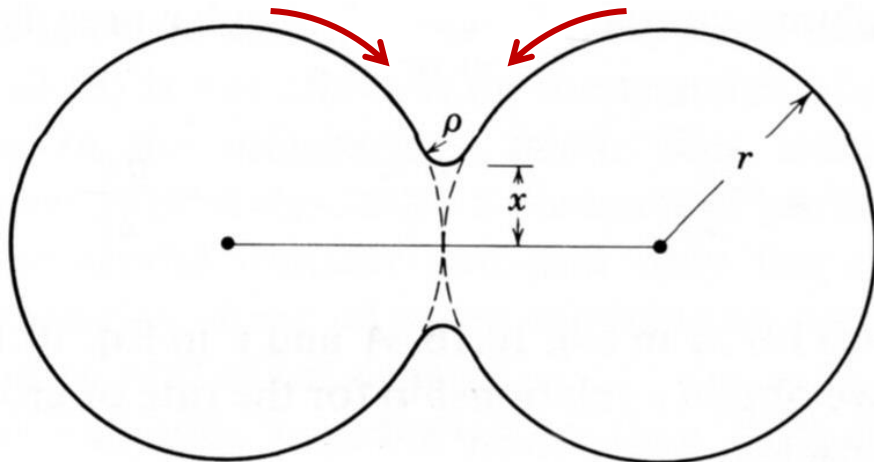
Start with a powder of the ceramic

- **Powder** - grind powder – **ball milling**

- compact and shape the powder
(sometimes with a binder)



- **Heat** in a furnace – powder particles join by surface diffusion below melting temperature ($\sim 80\% T_m$)



Sintering process

Flow of material from large curvature to small curvature by **surface diffusion** to reduce surface energy

Particulate forming processes

Sintering occurs during firing of a piece that has been powder pressed

powder particles coalesce and reduction of pore size

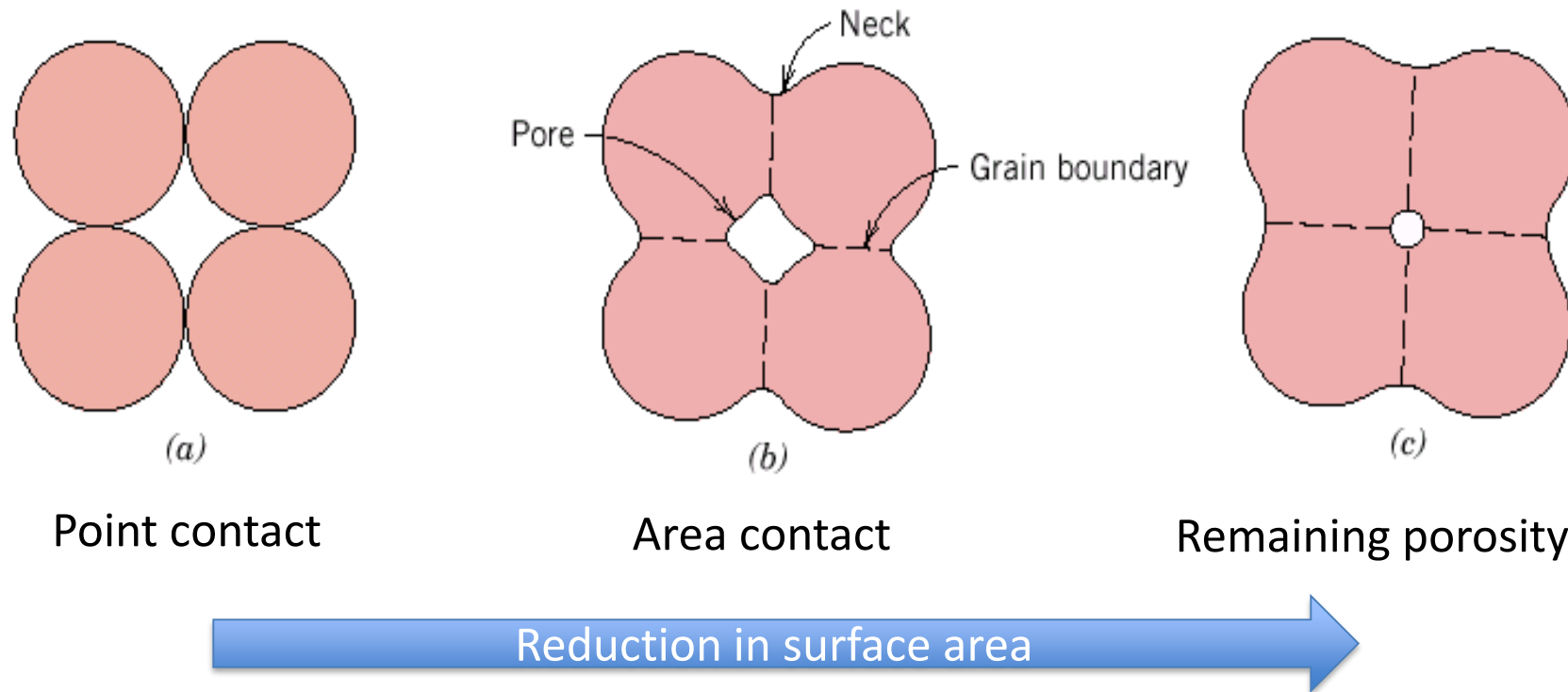
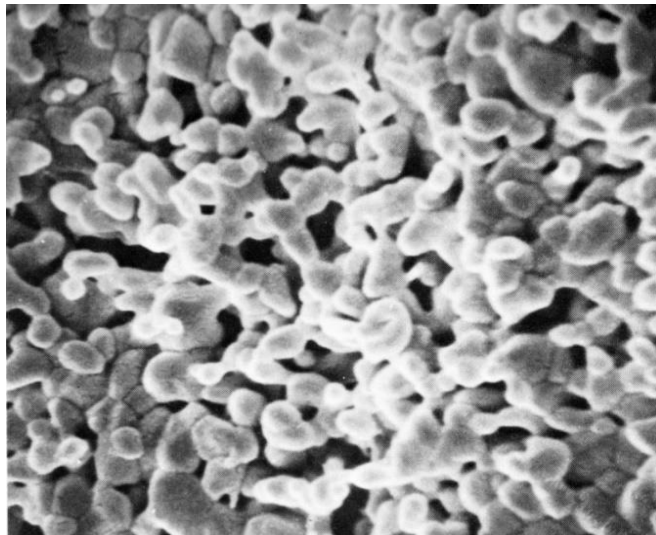


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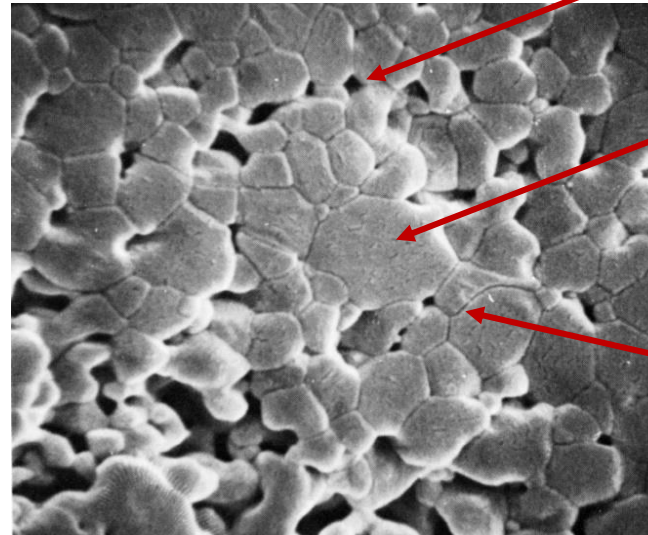
Particulate forming processes

Sintering ceramics to make products

- sinter at high temperature (below melting temperature $\sim > 80\% T_m$)
- surface diffusion joins powder particles together into a solid
- with pressure (= Hot pressing)



2 μm —



2 μm —

Al_2O_3 powder – after 2 ½ minutes and 6 minutes at 1700°C

Particulate forming processes

Fast sintering of ceramics

- **Spark Plasma Sintering (SPS)**
- Fast heating using pulsed DC electric current in graphite dies
- Heating to 2500°C at **1000°C/min**
- Pressure up to 1000MPa

Advantages of fast processing by SPS

- fabricate nanostructured ceramics
- produce textured microstructures
- preserve metastable microstructures



SPS machine at QMUL NanoForce Laboratory

Ceramic Fabrication Methods



GLASS
FORMING

PARTICULATE
FORMING

CEMENTATION

- Hardening of a paste – paste formed by mixing cement material with water
- Formation of rigid structures having varied and complex shapes
- Hardening process – hydration (complex chemical reactions involving water and cement particles)
- Examples:
 - Portland cement
 - Dental crowns



Ceramics processing:

High temperature – high energy processes

- Glass – reduce the melting or softening temperature
- Particle forming – process below melting temperature
 - Sintering = surface diffusion to consolidate
 - Flaws left in material will be important for reliability
- Cementation – Hardening of paste formed by mixing cement material with water

What can you do now?

- Recognise the different types of ceramic materials ‘in-organic and non-metallic materials’
- Use crystal site selection rules to predict ceramic crystal structures and crystal defects in ceramics
- Understand how ceramics change their behaviour with phase and crystal polymorph types (including amorphous phases)
- Understand how to process ceramics
- Select processing methods and heat treatments to achieve the desired performance of ceramics